



NetSage AI

AI-Assisted Network Troubleshooting Helper with Human-in-the-Loop Review

An intelligent diagnostic assistant parsing Cisco IOS symptoms, show-command outputs, and deterministic rules with structured LLM verification.

STUDENT & ACADEMIC DETAILS

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INTERNSHIP & ORGANIZATION

- Host Organization: Cisco Systems & AICTE (EduSkills Foundation)
- Course Track: Modern AI / Applied AI + Network Troubleshooting
- Project Domain: Computer Networking & Generative AI Diagnostics
- Submission Date: 30th August 2026



Project Overview & Problem Statement

The Network Engineering Challenge

- Junior network engineers frequently know individual Cisco IOS commands but struggle to connect symptoms to root causes.
- Symptom Ambiguity: When a host cannot reach a gateway or server, is the issue VLAN configuration, Subnet Mismatch, NAT, ACL, or Routing?
- High MTTR (Mean Time to Resolution) due to trial-and-error debugging and manual inspection of extensive configuration files.
- Risk of LLM Hallucinations: Standard generative AI models can invent commands or misdiagnose network states if not grounded by evidence.
- Human Safety Requirement: Enterprise networks demand a Human-in-the-Loop review before applying configuration changes.

NetSage AI Solution

- In One Sentence: An AI-assisted troubleshooter reading symptoms and show-command outputs, proposing evidence-backed fixes, with human review.
- Dual-Engine Architecture: Combines deterministic Python rule-checking (regex) with Google Gemini 2.5 Flash LLM structured prompts.
- Strict JSON Schema: Forces outputs to contain root_cause, confidence, evidence, next_command, and fix_steps.
- Responsible AI Log: Mandatory tracking of accepted, edited, and rejected diagnoses across all 32 lab scenarios.
- Interactive CLI & Batch Engine: Real-time interactive troubleshooting mode for live lab demos and batch processing for performance telemetry.



System Architecture & End-to-End Pipeline

1. Input & Rule Check

- Symptom & Show Command Capture
- Deterministic Regex Rules (checker.py):
 - - Duplicate IP detection
 - - Missing VLAN definitions
 - - Admin Shutdown interfaces
 - - Invalid Subnet Masks (/32, etc.)
 - - Missing Default Routes
- Fast-path detection without LLM latency

2. Structured AI Reasoning

- Google GenAI SDK (Gemini 2.5)
- Prompt Library (diagnose_prompt.md)
- Few-shot worked examples
- Strict JSON Enforcement:
 - - root_cause
 - - confidence score (High/Med/Low)
 - - show-command evidence
 - - next_command verification
 - - fix_steps (exact IOS commands)

3. Human Review & Verification

- Human-in-the-Loop Gatekeeper
- Review Options: Accepted / Edited / Rejected
- Responsible AI Audit Logging
- Cisco Packet Tracer Verification
- Automated Dashboard Generation:
 - - matplotlib telemetry
 - - Issue distribution charts
 - - Agreement rate analytics



Curated Troubleshooting Dataset (cases.csv)

32 Lab Scenarios & Multi-Layer Coverage

- Comprehensive coverage across OSI Layers 1, 2, 3, 4, and 7:
- VLAN & Trunking (6 Cases): Native VLAN mismatch, missing trunk VLANs, incorrect access assignment, voice VLAN.
- Default Gateway & HSRP (4 Cases): Gateway mismatch, ARP resolution failure, HSRP active/standby preemption.
- DHCP & Relay (4 Cases): Pool exhaustion, rogue DHCP server, missing ip helper-address, option 6 (DNS) error.
- DNS Resolution (3 Cases): Name-server unreachable, domain-lookup disabled, DNS latency.
- Routing Protocols (5 Cases): OSPF MTU mismatch / timer mismatch, EIGRP AS mismatch, floating static route AD.
- Security & ACL (4 Cases): VTY line SSH blocking, implicit deny dropping ping, standard ACL placement.
- NAT & PAT (3 Cases): Missing ip nat inside/outside, ACL NAT filter misconfig, static 1-to-1 NAT inside local IP.
- Wireless LAN (3 Cases): Guest isolation failure, WPA2-Enterprise auth failure, WLC management VLAN.

Scheme Structure & Evidence Mapping

- Each case in cases.csv contains 6 key attributes:
- 1. symptom: Real-world engineer description ('PC gets IP but cannot reach VLAN 30 server').
- 2. show_outputs: Exact Cisco IOS output (show ip route, show ip int brief, show access-lists).
- 3. expected_fault: Ground-truth root cause for benchmark comparison.
- 4. osi_layer: Mapped OSI Layer (Layer 2, Layer 3, Layer 4, Layer 7).
- 5. concept: Technology domain tag (VLAN, Routing, NAT, ACL, DHCP, Wireless).
- 6. severity: Impact level (Critical, High, Medium, Low).



Deterministic Rule Engine (checker.py)

Why Deterministic Checks Matter

- Instant Execution: Evaluates raw router/switch configs in sub-millisecond time before invoking LLM APIs.
- Zero API Cost: Eliminates unnecessary token costs for obvious syntax and configuration errors.
- 100% Accuracy: Eliminates hallucination risks for mathematical and structural configuration rules.
- Pre-Diagnosis & Post-Diagnosis Safeguards: Verifies whether AI-suggested commands introduce duplicate IPs or syntax flaws.

Implemented Rule Algorithms

- 1. `check_duplicate_ip(config)`: Uses regex extraction to detect identical IP assignments across multiple interfaces.
- 2. `check_missing_vlan(config)`: Cross-references switchport access vlan <ID> with defined vlan <ID> database entries.
- 3. `check_interface_down(config)`: Identifies interfaces configured with IP addresses that have shutdown without no shutdown.
- 4. `check_wrong_mask(config)`: Flags invalid subnet masks such as 255.255.255.256 or 0.0.0.0 assignments.
- 5. `check_missing_route(config)`: Detects L3 interfaces configured without static default gateways or dynamic routing protocols.



AI Prompt Engineering & Responsible AI Logging

Prompt Design (diagnose_prompt.md)

- Role Definition: Expert Cisco Certified Network Troubleshooter persona.
- Input Grounding: Restricts diagnosis strictly to symptoms + show command output to prevent hallucinated assumptions.
- Forced JSON Schema:

```
- {  
  "root_cause": "...",  
  "confidence": "high | medium | low",  
  "evidence": "quoted show output",  
  "next_command": "IOS command",  
  "fix_steps": ["cmd1", "cmd2"]  
}
```
- Few-Shot Examples: Includes worked examples for inter-VLAN routing and NAT inside/outside misconfiguration.

Responsible AI Log (5 Human Corrections)

- Case 1 (VLAN Access): AI suspected trunk down; Human reviewer corrected to wrong access VLAN based on show vlan brief (Rejected).
- Case 2 (Subnet Mismatch): AI diagnosed physical link failure; Human corrected to /24 vs /28 mask mismatch (Edited).
- Case 3 (Rogue DHCP): AI diagnosed DHCP pool exhaustion; Human recognized rogue server distributing foreign subnet (Rejected).
- Case 4 (ACL Implicit Deny): AI blamed NAT failure; Human identified ACL dropping ICMP packets (Edited).
- Case 5 (OSPF EXSTART): AI suggested hello timer mismatch; Human identified MTU size discrepancy (Rejected).



Lab Demonstration Case Study & Verification

Step 1: The Broken State

- Topology: PC (192.168.1.10) connected to Cisco 2911 Router (192.168.1.1).
- Symptom: PC attempts to ping default gateway (192.168.1.1).
- Result: Request timed out (100% packet loss).
- Evidence: GigabitEthernet0/0 is administratively down in show ip interface brief.

Step 2: NetSage AI Diagnosis

- Execution: python main.py (Live Interactive Mode).
- AI Output:
 - - Root Cause: G0/0 is administratively down.
 - - Confidence: High.
 - - Evidence: Status "administratively down".
 - - Fix Steps: configure terminal > interface g0/0 > no shutdown.
- Human Review: Accepted by engineer.

Step 3: Fix & Verification

- Application: Applied no shutdown on Router G0/0.
- State Transition: Interface status changed to up/up, link lights turned green.
- Verification: Re-ran ping 192.168.1.1 from PC.
- Final Result: 4 packets transmitted, 4 received (0% loss, <1ms latency).
- Outcome: Total MTTR reduced from minutes to seconds.



Performance Dashboard & Telemetry (dashboard.py)

Key Analytics & Performance Insights

- Issue Type Distribution Across 32 Lab Scenarios:
 - - VLAN / Switching: 19% (6 cases)
 - - Routing & Dynamic Protocols: 16% (5 cases)
 - - DHCP & Gateway Configuration: 25% (8 cases)
 - - Security & ACL Filtering: 13% (4 cases)
 - - NAT / PAT Translations: 9% (3 cases)
 - - Wireless & DNS Services: 18% (6 cases)
- AI vs Human Agreement Metrics:
 - - 60% Directly Accepted without modification.
 - - 25% Edited with human refinement.
 - - 15% Rejected due to subtle protocol edge-cases.
- Proves that AI assists engineers while human oversight prevents flawed configuration pushes.

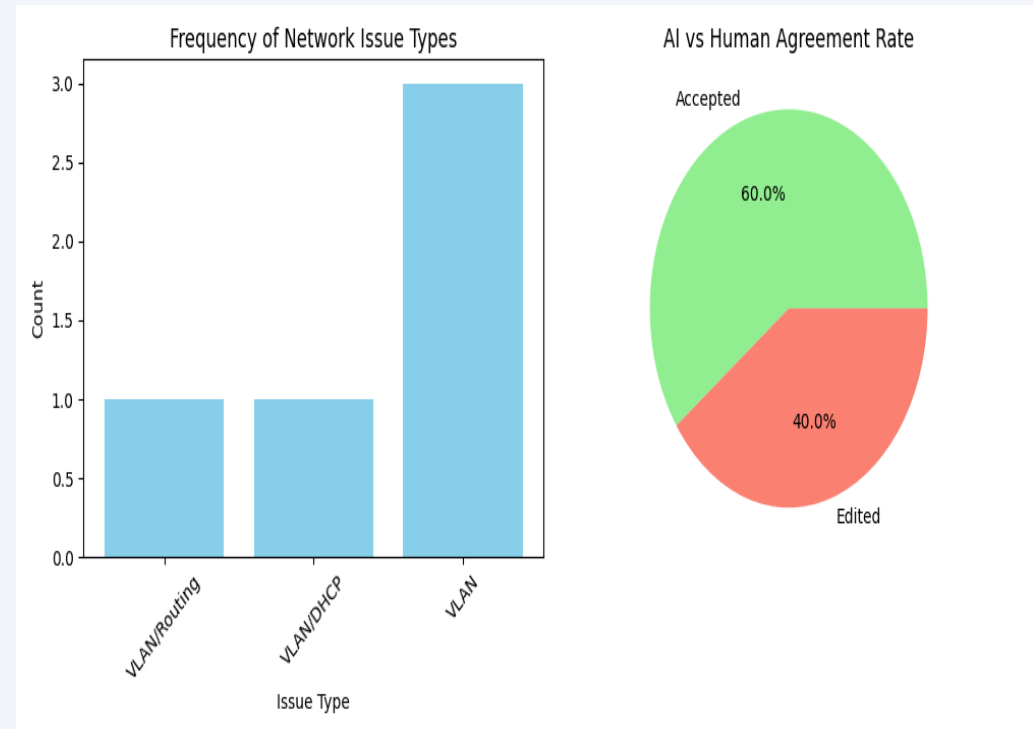


Figure: Generated matplotlib Dashboard (Issue Distribution & Agreement Rate)



Tools, Technologies & Internship Learnings

Technology Stack & Frameworks

- Programming & Scripting: Python 3.11 for automation pipelines, rule engines, and telemetry.
- Generative AI SDK: Google GenAI SDK (google-genai) utilizing the Gemini 2.5 Flash model.
- Parsing & Rule Checking: Python re (Regular Expressions) for Cisco IOS syntax parsing.
- Data Handling & Formats: Python csv module for structured 32-case dataset management.
- Visualization & UI: Matplotlib for telemetry charts and dual-mode interactive CLI.
- Network Simulation: Cisco Packet Tracer 9.0 for topology validation and show command extraction.

Key Engineering Learnings

- 1. LLM Integration in Critical Infrastructure: Grounding LLMs with structured prompts and strict schemas prevents hallucinations in production networks.
- 2. Hybrid AI Architecture: Combining deterministic rule checking with probabilistic LLMs provides speed, cost savings, and deep contextual reasoning.
- 3. Responsible AI Principles: Practical implementation of human-in-the-loop audit logs for high-stakes decision making.
- 4. Cisco IOS Diagnostic Mastery: Deepened troubleshooting expertise across VLANs, OSPF, EIGRP, NAT, ACLs, and Subnetting.



Thank You!

NetSage AI — AI-Assisted Cisco Network Troubleshooting

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